EXERCISE 1

(a) Assumptions

1) CustomerNumber uniquely identifies a customer (CustomerNumber -> CustomerName, CustomerType).
2) An order is uniquely identified by (CustomerNumber, Date, Time). A customer does not place two different orders at the exact same date+time.
3) A single order can contain many parts (repeating group is the part lines).
4) PartNumber uniquely identifies a part (PartNumber -> PartName, PartType, CageCode, UnitPrice).
5) EmployeeName is recorded directly on the order (no EmployeeID attribute exists, so we do not create an Employee table).

(b) Normalization process

<u>UNF -> 1NF:</u>

- Problem: repeating group of parts on the order form.
- Fix: store one row per ordered part (atomic values). PK becomes (CustomerNumber, Date, Time, PartNumber).

Exercise 1 - 1NF (single table; one row per part on the order)

OrderForm1NF

CustomerNumbe	CustomerName	CustomerType	**Date**	**Time**	EmployeeName	**PartNumber**	PartName	PartType	CageCode	QuantityOrdered	UnitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D.Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D.Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D.Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D.Harrison	10657	Float Assembly	Plumbing	G413	5	10

<u>1NF -> 2NF:</u>

- Problem: composite PK causes partial dependencies:
  CustomerNumber -> CustomerName, CustomerType
  (CustomerNumber, Date, Time) -> EmployeeName
  PartNumber -> PartName, PartType, CageCode, UnitPrice
- Fix: split into Customer, Order, Part, and OrderLine; keep QuantityOrdered in OrderLine because it depends on the full PK.

Exercise 1 - 3NF Relations (also satisfies 2NF)

Customer

CustomerNumber	CustomerName	CustomerType
HG54587	Jeff Peterson	Consumer

Order

CustomerNumber	Date	Time	EmployeeName
HG54587	7/1/2024	10:30am	D.Harrison

OrderLine

CustomerNumber	Date	Time	PartNumber	QuantityOrdered
HG54587	7/1/2024	10:30am	10654	4
HG54587	7/1/2024	10:30am	10456	3
HG54587	7/1/2024	10:30am	10776	7
HG54587	7/1/2024	10:30am	10657	5

Part

PartNumbe	PartName	PartType	CageCode	UnitPrice
10654	Float Contro	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assemb	Plumbing	G413	9
10657	Float Assemb	Plumbing	G413	10

<u>2NF -> 3NF:</u>
- Check for transitive dependency among non-key attributes: none given the attributes available.
- Therefore, 3NF relations are the same set as 2NF relations.

(c)keys in each step
<u>1NF:</u>
- OrderForm1NF PK = (CustomerNumber, Date, Time, PartNumber)

<u>3NF (same as 2NF):</u>
- Customer PK = CustomerNumber
- Order PK = (CustomerNumber, Date, Time); FK: CustomerNumber -> Customer(CustomerNumber)
- Part PK = PartNumber
- OrderLine PK = (CustomerNumber, Date, Time, PartNumber)
  FKs:
  (CustomerNumber, Date, Time) -> Order(CustomerNumber, Date, Time)
  PartNumber -> Part(PartNumber)

(d) 3NF relation names
Customer, Order, Part, OrderLine

EXERCISE 2

(a) Assumptions
1) staffNo uniquely identifies a therapist (staffNo -> therapistName).

2) patNo uniquely identifies a patient (patNo -> patName).
3) A therapist can see at most one patient at a given (appointmentDate, appointmentTime).
Therefore an appointment is uniquely identified by (staffNo, appointmentDate, appointmentTime).
4) Therapists may work at multiple branches, but on any given day each therapist works at exactly one branch:
(staffNo, appointmentDate) -> branchNo.

(b) Normalization process

The start table is already 1NF (atomic), but has dependencies that break 2NF/3NF.

Exercise 2 - 1NF (single table)

TherapistAppointment1NF

staffNo	therapistName	patNo	patName	appointmentDate	appointmentTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

<u>1NF -> 2NF:</u>
- Composite PK (staffNo, appointmentDate, appointmentTime) creates partial dependency:
  staffNo -> therapistName
  (staffNo, appointmentDate) -> branchNo
- Fix: create Therapist(staffNo, therapistName) and TherapistDayBranch(staffNo, appointmentDate, branchNo).
- Keep Appointment with PK (staffNo, appointmentDate, appointmentTime) and store patNo there.

Exercise 2 - 2NF Relations

Therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

TherapistDayBranch

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

Appointment

staffNo	**appointmentDate**	**appointmentTime**	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

2NF -> 3NF:

- Transitive dependency in Appointment data:
  patNo -> patName (non-key determines non-key)
- Fix: create Patient(patNo, patName). Keep only patNo in Appointment.

Exercise 2 - 3NF Relations

Therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

Patient

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

TherapistDayBranch

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

Appointment

staffNo	***appointmentDate***	**appointmentTime**	*patNo*
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

(c) keys

1NF:

- TherapistAppointment1NF PK = (staffNo, appointmentDate, appointmentTime)

2NF:

- Therapist PK = staffNo

- TherapistDayBranch PK = (staffNo, appointmentDate)
- Appointment PK = (staffNo, appointmentDate, appointmentTime)

<u>3NF:</u>
- Therapist PK = staffNo
- Patient PK = patNo
- TherapistDayBranch PK = (staffNo, appointmentDate)
  FK: staffNo -> Therapist(staffNo)
- Appointment PK = (staffNo, appointmentDate, appointmentTime)
  FKs:
  staffNo -> Therapist(staffNo)
  patNo -> Patient(patNo)
  (staffNo, appointmentDate) -> TherapistDayBranch(staffNo, appointmentDate)

(d) 3NF relation names
Therapist, Patient, TherapistDayBranch, Appointment

EXERCISE 3

(a) Assumptions
1) eNo uniquely identifies an employee (eNo -> eName).
2) contractNo uniquely identifies a contract.
3) Each contract applies to exactly one event: contractNo -> eventNo.
4) Each event has one location: eventNo -> eventLoc.
5) hours worked is specific to an employee on a contract: (eNo, contractNo) -> hours.

(b) Normalization process
Start table is 1NF (atomic) but has partial + transitive dependencies.

Exercise 3 - 1NF (single table)

EmployeeContractHours1NF

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

<u>1NF -> 2NF:</u>
- Composite PK (eNo, contractNo) causes partial dependencies:

eNo -> eName
contractNo -> eventNo (and eventLoc information linked to contract/event)
- Fix: create Employee(eNo, eName) and Contract(contractNo, eventNo, eventLoc), plus WorkAssignment(eNo, contractNo, hours).

Exercise 3 - 2NF Relations

Employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

Contract

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

WorkAssignment

eNo	**contractNo**	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

<u>2NF -> 3NF:</u>
- Transitive dependency inside Contract:
contractNo -> eventNo and eventNo -> eventLoc
- Fix: split into Contract(contractNo, eventNo) and Event(eventNo, eventLoc).

Exercise 3 - 3NF Relations

Employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

Contract

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

Event

eventNo	eventLoc
H25	Queens
H4	Yonkers

WorkAssignment

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

(c)keys
<u>1NF</u>: PK = (eNo, contractNo)

2NF
- Employee PK = eNo
- Contract PK = contractNo
- WorkAssignment PK = (eNo, contractNo)

3NF:
- Employee PK = eNo
- Event PK = eventNo
- Contract PK = contractNo
  FK: eventNo -> Event(eventNo)
- WorkAssignment PK = (eNo, contractNo)
  FKs:
  eNo -> Employee(eNo)
  contractNo -> Contract(contractNo)

(d) 3NF relation names
Employee, Contract, Event, WorkAssignment